Name: _____

Class: _____

Date: _____ **Mr. Rawson • GCSE Physics**

Writing Up a Practical

Five moves, one or two sentences each. Every sentence carries a number or a variable name from **your own** data, and a *because* wherever you make a claim.

Model paragraph: a different practical (stretching a spring)

I changed the mass hung on the spring and measured its extension, keeping the same spring and the same ruler. As the mass increases, the extension increases. The relationship is direct, and the correlation is strong because every point sits close to the trendline; run 4 is the one point that sits clearly away from it.

The move	Sentence starters	In RP19, look at
1. Describe the pattern. State the type of relationship, how strong the correlation is, and name any anomaly.	As _____ increases, _____ increases / decreases. The relationship is _____ (direct / inverse), with _____ correlation because the points _____. Run _____ sits clearly away from the trendline.	Graph A and Graph B on the spreadsheet, and the run numbers in your two tables.
2. Give the numbers. Quote your trendline and the theoretical value, with units.	My trendline is _____. The theory predicts _____. The percent error is _____%.	The yellow box under each graph, and the percent error formula in the blue box.
3. Link to the law. Rearrange the formula until its shape matches your graph.	Rearranging _____ gives _____, which is the shape of a _____ graph. The gradient / coefficient should equal _____ because _____.	$\Sigma F = m \cdot a$, rearranged for a. Part A compares with $1 \div M$; Part B with the 0.49 N weight.
4. Judge the accuracy. Place your percent error in its band, then say whether the data supports the law.	My error of _____% is below / between / above the 5% and 10% limits, so the data is _____. This supports / does not support _____ because _____.	The three accuracy bands in the blue box on the Analysis page.
5. Improve it. Name one source of error, say which way it moves the result, and give a fix.	_____ made the measured _____ too high / too low, because _____. Doing _____ would reduce this.	The intercept on Graph A, where you placed the Linear Fit, and the friction in the track.

Words to use

- **Directly proportional** means that when one quantity doubles the other doubles, and the graph is a straight line through the origin.
- **Inversely proportional** means that when one quantity doubles the other halves, and the graph is a curve that flattens out.
- **Gradient** is the change in y divided by the change in x, and it carries a unit.
- **Correlation** describes how closely the points follow the trendline: perfect, strong or weak.
- **Anomaly** is a point that sits clearly away from the pattern the other points make.

Read it back. Underline every number in your write-up and check that each has a unit and comes from your table or your trendline. Ring every variable you name and check it is written in full ("total mass", not "it"). Find each claim and check that a *because* follows it.